



Oral rad 25-26

Good Morning

ADVANCED IMAGING MODALITIES

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- **Lecture learning outcomes**

By the end of the lecture students should be able to:

- Enumerate the advanced imaging techniques of the maxillofacial tissues.
- Describe the working principle of CT, MRI, PET & Ultrasound scans.
- Enumerate the indications, contraindications, advantages and disadvantages of CT, MRI, PET and Ultrasound scans.

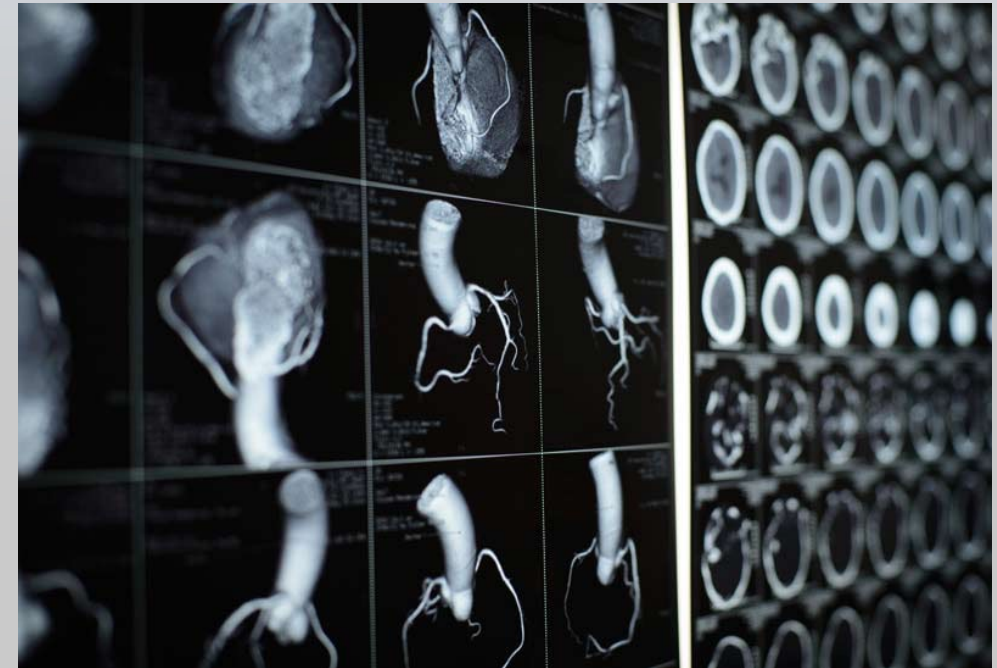
ADVANCED IMAGING MODALITIES

TOMOGRAPHY

- TOMO- “Slice”
- GRAPHIA- “describing”/ “writing”

Tomography is a radiographic technique that allows the imaging of one layer or section of the body while blurring images from structures in other planes.

- Computed tomographic (CT) scanning
- Magnetic resonance (MR) imaging.
- Ultrasonography,
- Nuclear medicine- Positron emission tomographic (pet) imaging
- Cone beam computed tomography (CBCT)





COMPUTED TOMOGRAPHY [CT]

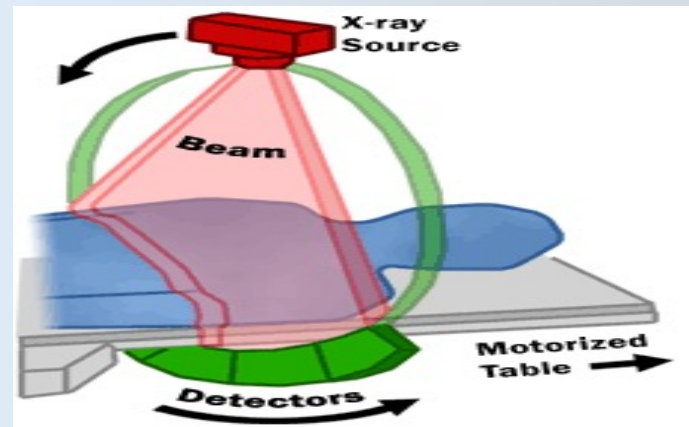
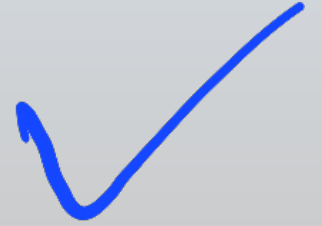
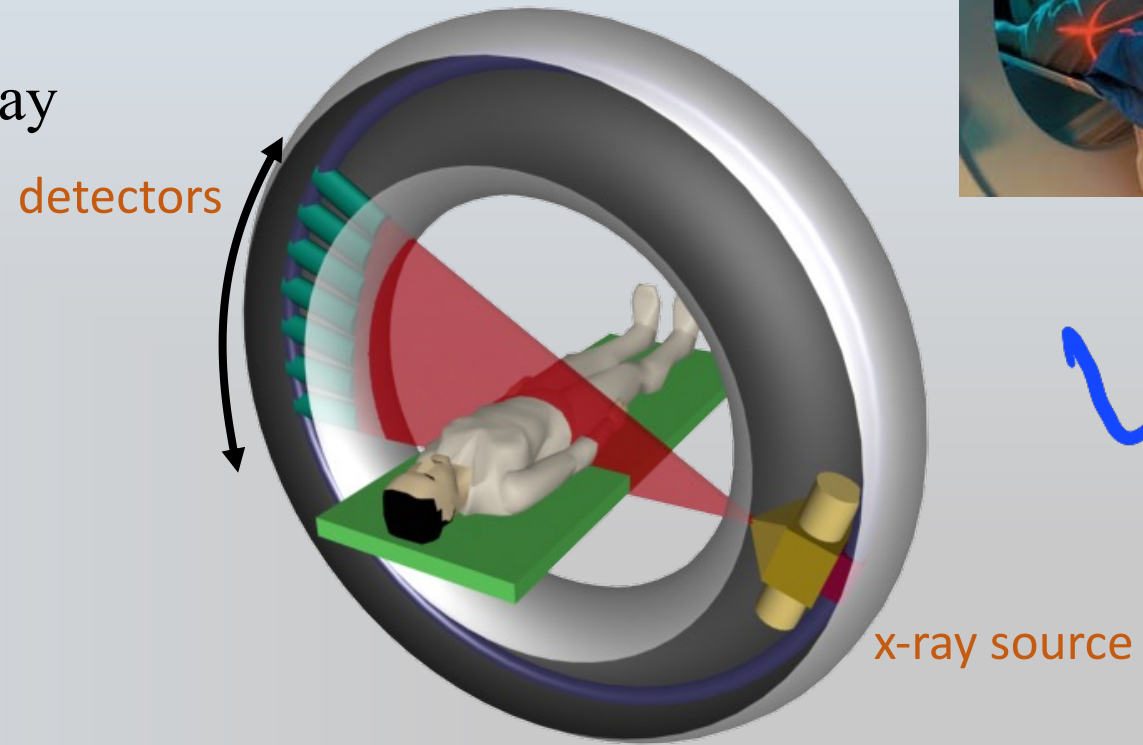
The CT scanner was introduced by **Sir Godfrey N. Hounsfield** in 1967

- All CT scanners consist of an x ray rotating gantry.

1.The gantry houses

- a. X-ray Tube
- b.Detectors

2.Computer Imaging Processor



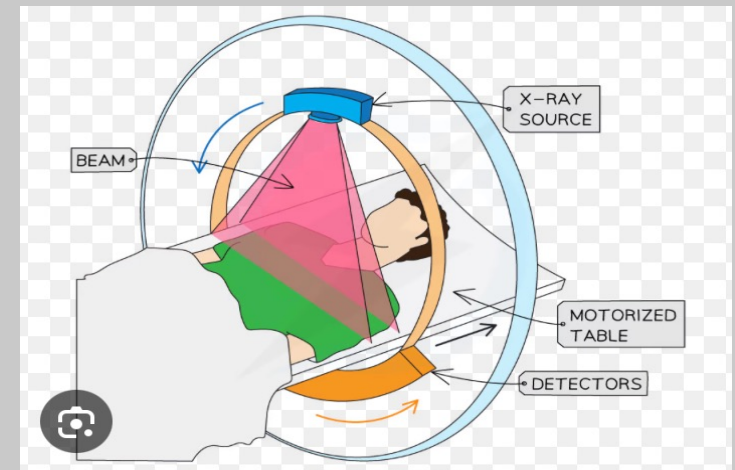
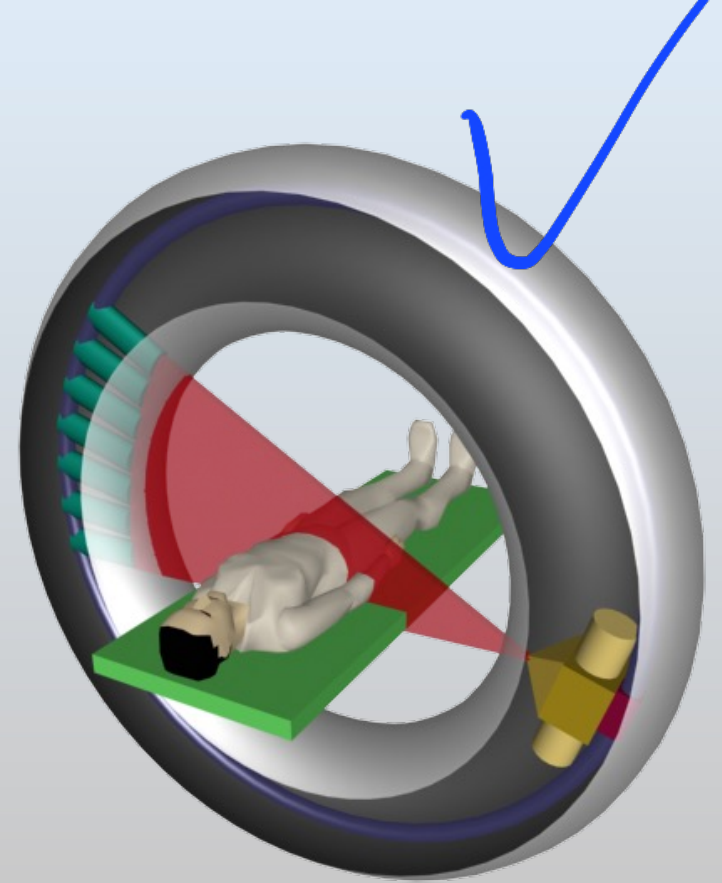
The patient's body to be interposed between the x-ray source and the detectors

1.The gantry houses

- a. X-ray Tube, b.Detectors

a. X-RAY TUBE :

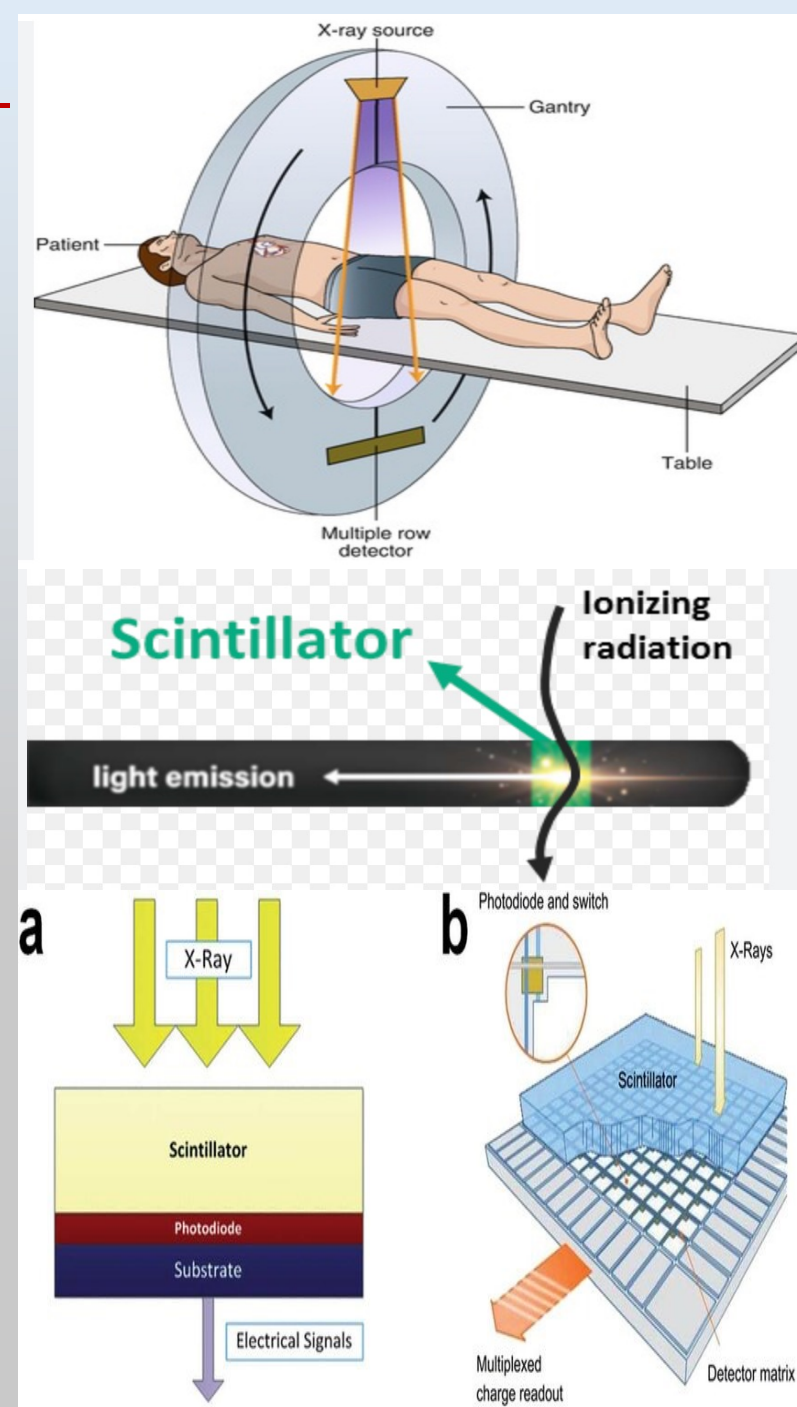
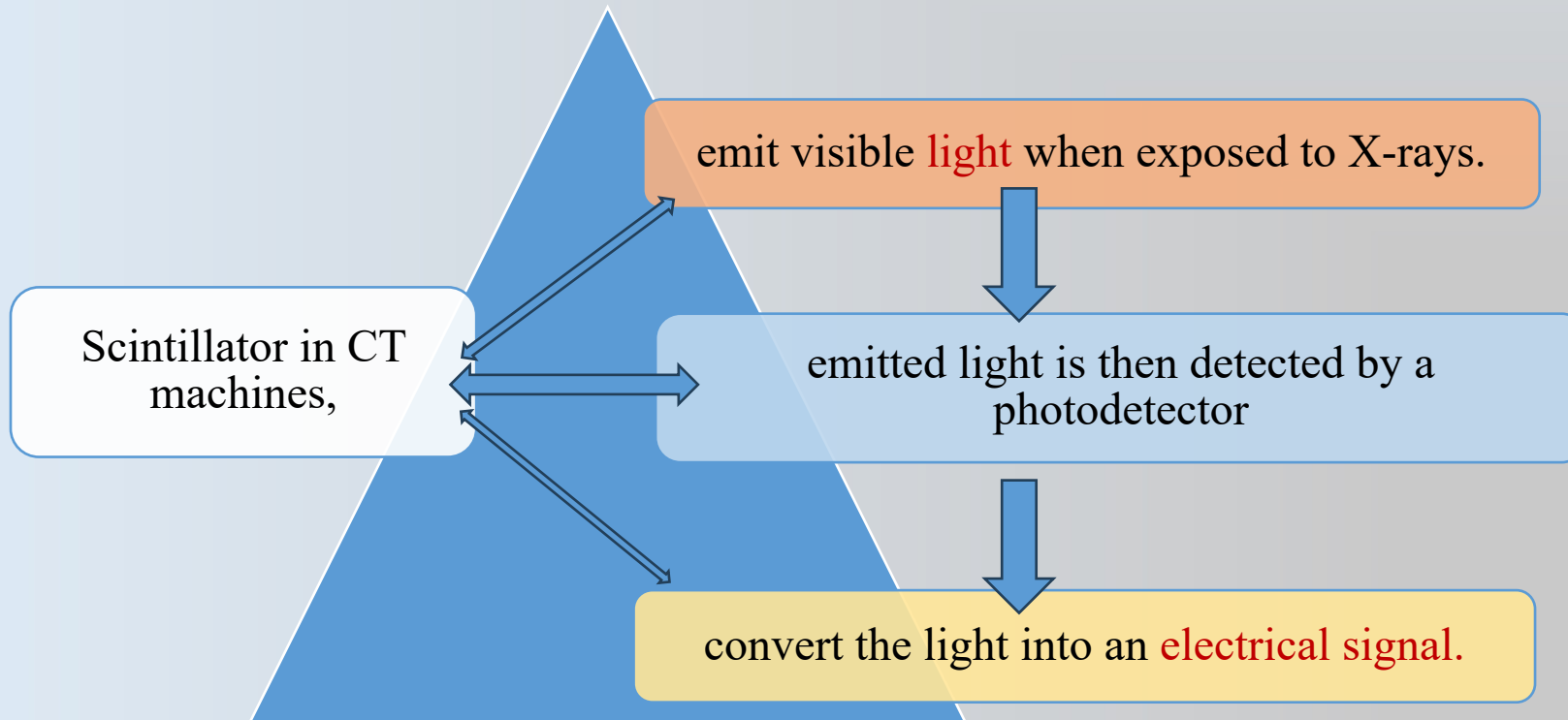
- Rotating anodes .
- They operate at typically 120 kVp (80-120)
- 200 to 800 mA.
- Thin fan beam collimated



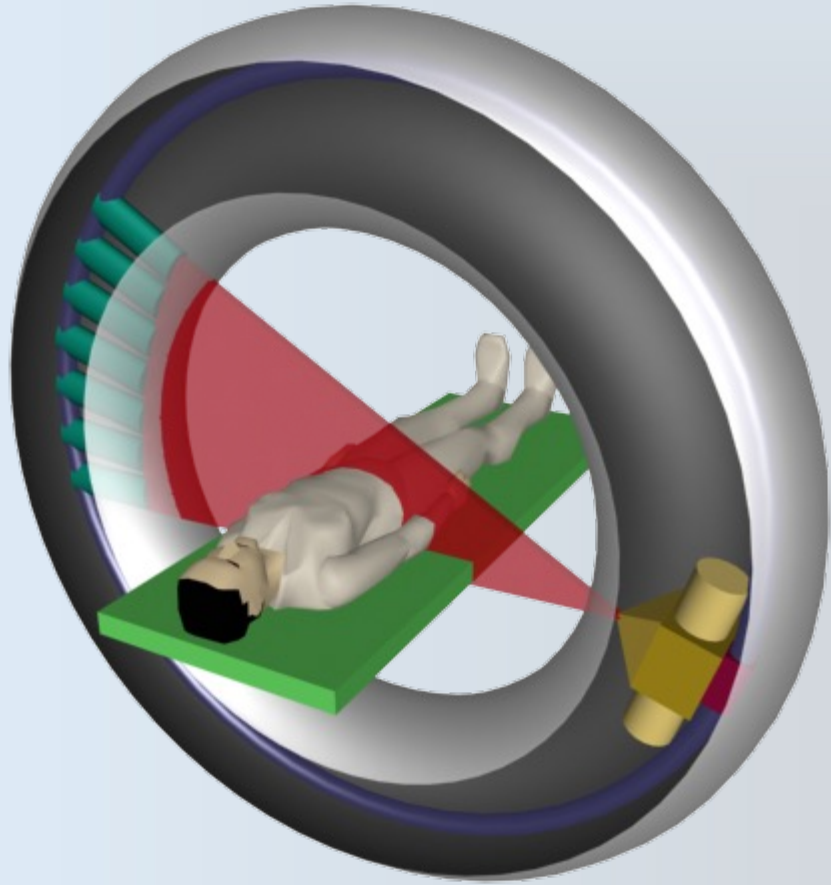
The x-ray beam exiting the patient is captured by an array of solid-state detectors.

b.DETECTORS-

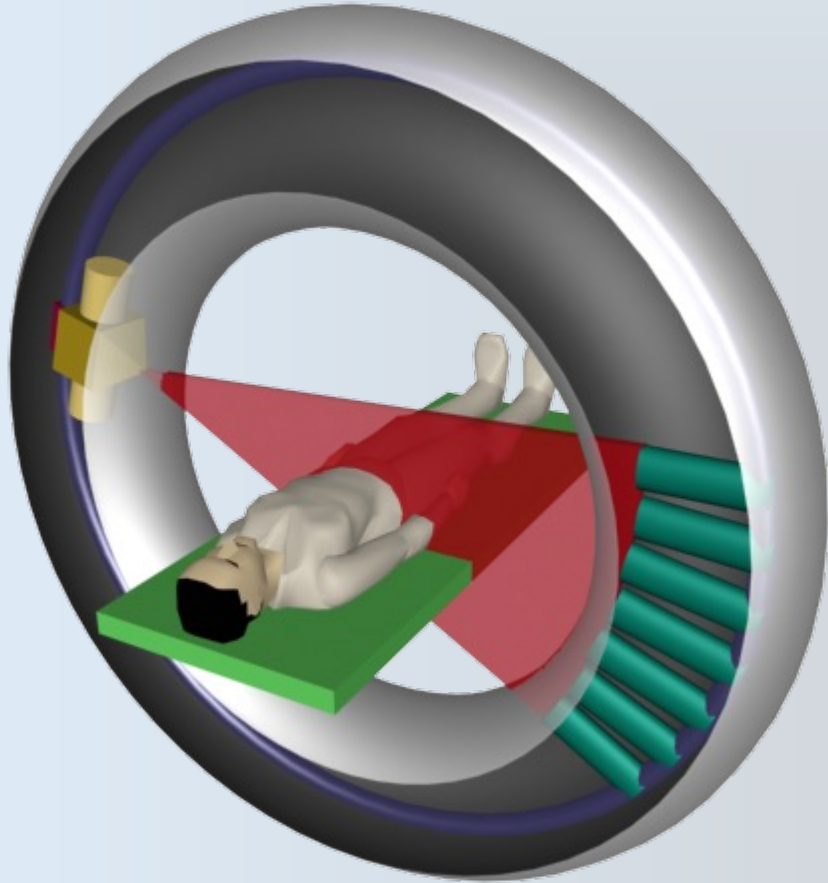
Scintillation crystals- $\text{Gd}_2\text{S}_2\text{O}$ (Gadolinium oxysulfide)



A series of exposures are made
over an arc of 180 or 360 degrees.



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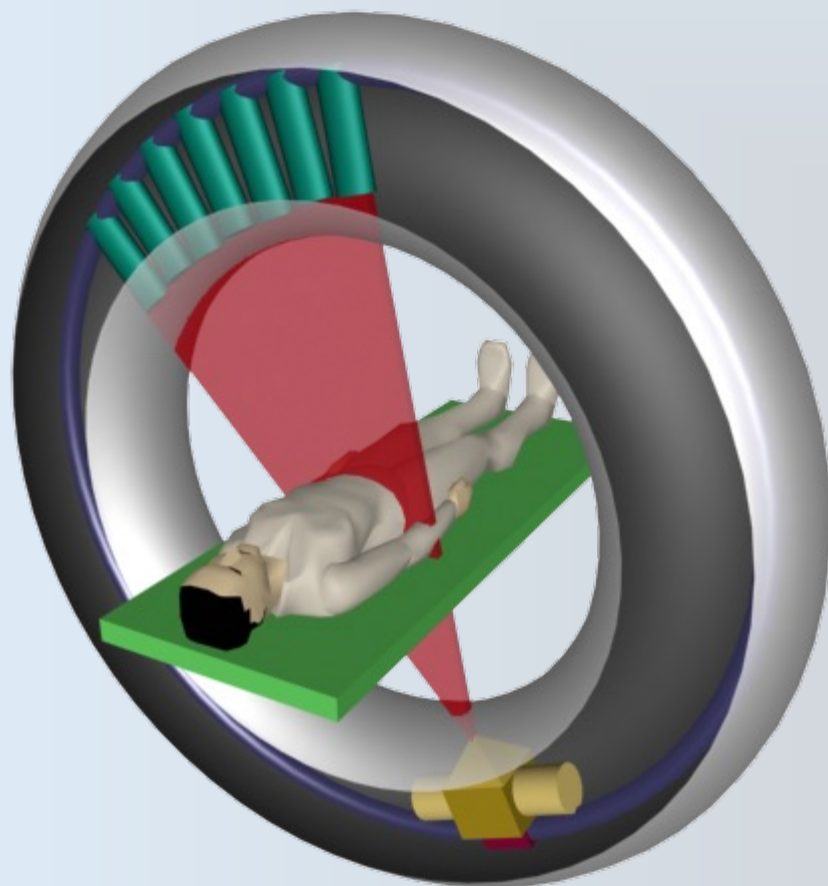


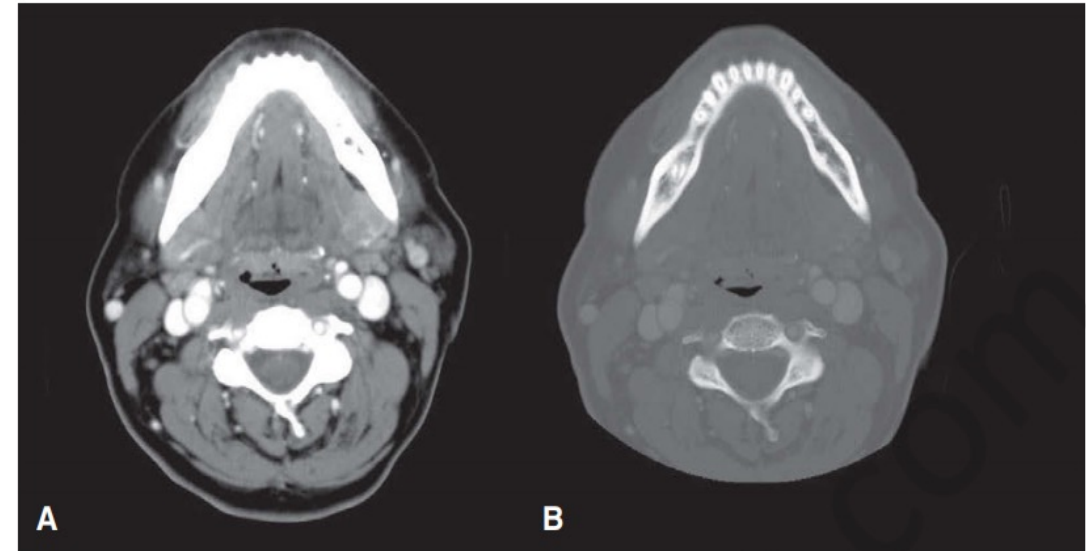
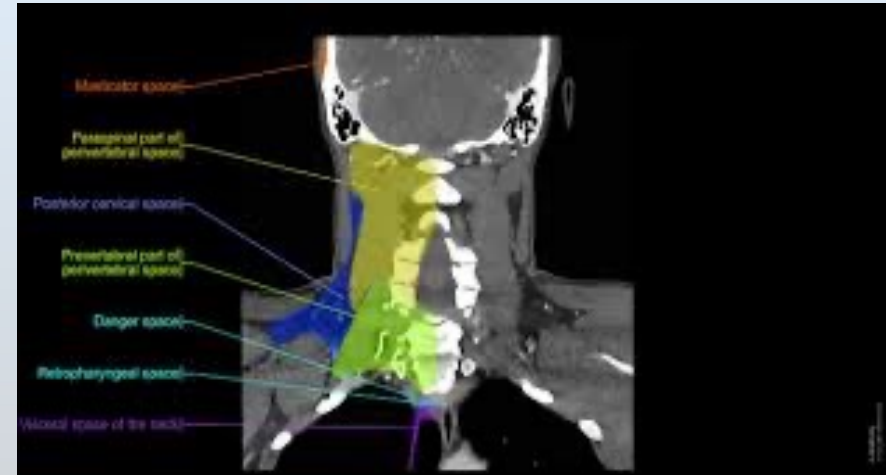
TABLE 14-1 Typical Hounsfield Units for Air and Tissues

Tissue	Hounsfield Units (CT Numbers)
Bone	+400 to +1000
Soft tissue	+40 to +80
Water	0
Fat	-60 to -100
Lung	-400 to -600
Air	-1000

- The basic principle of CT imaging is that -different tissues absorb X-rays differently.
- **Dense tissues (e.g., bone)** absorb more X-rays.
- **Less dense tissues (e.g., air in lungs)** absorb fewer X-rays.
- The CT scanner assigns a “CT number” (Hounsfield unit) to each tissue:
- **+1000** for dense bone
- **0** for water
- **-1000** for air
- These numbers appear as gray shades, creating detailed 2D and 3D images

Contrast CT imaging –

- Contrast agents are substances used to improve visualization of Structures, enhance soft tissue and vascular image .
- Contrast agent- Iodine(Intra Venous).
- Toxic to patients with kidney disease , asthmatic patients.



Contrast enhanced CT

Indications:

➤ In Maxillofacial Region:

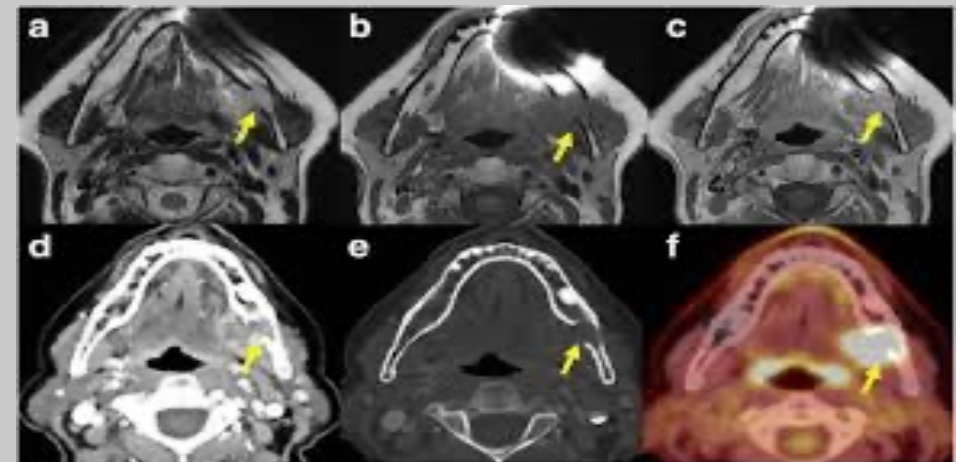
1)Diagnoses and assess :

- Extent of Infections,
- Osteomyelitis,
- Cysts,
- Tumors (Benign and malignant).

3)Assists in Trauma cases and

4)Craniofacial reconstruction.

5)Creates life-sized models for surgeries, dental implants, and prostheses

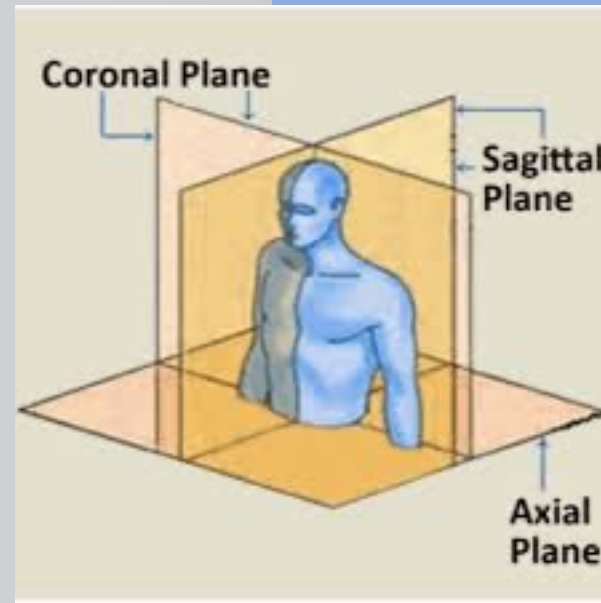
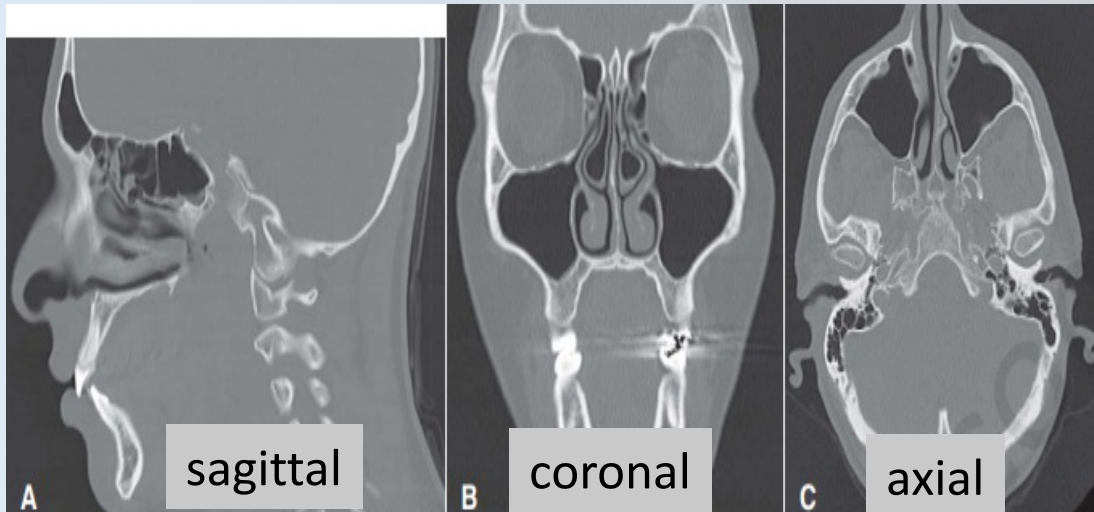


ADVANTAGES

- Ideal for hard tissue imaging.
- Eliminates overlap from surrounding areas.
- Allows image manipulation.
- Enables multiplanar views (axial, coronal, sagittal).

DISADVANTAGES

- High radiation dose
- Metallic restorations may produce artifacts
- Less effective soft tissue contrast compared to MRI





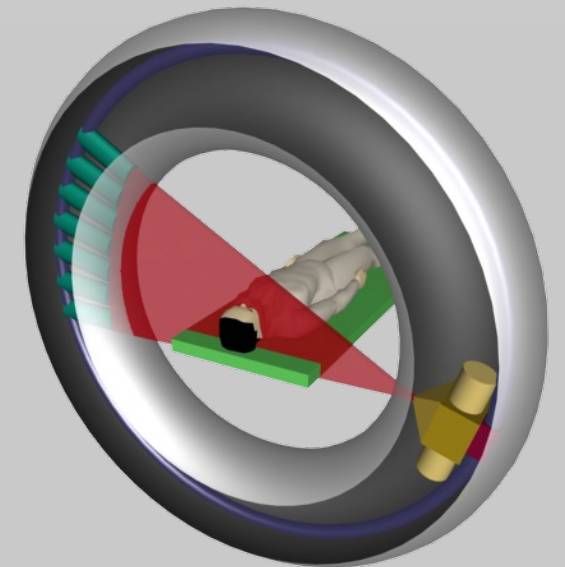
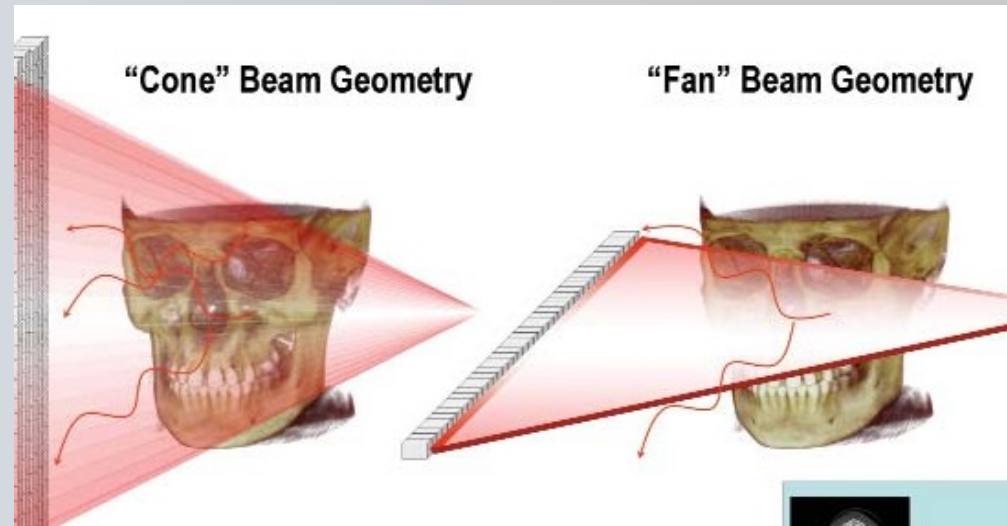
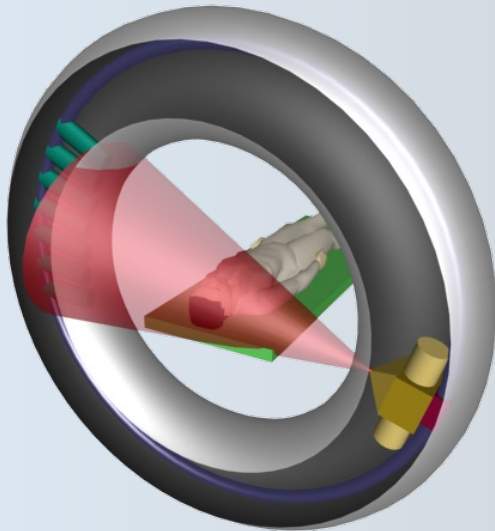
CONE BEAM COMPUTED TOMOGRAPHY (CBCT)

CBCT is a type of specialized X-ray used primarily in dental and maxillofacial imaging.

CBCT uses a **cone-shaped X-ray beam** to capture a large volume in a **single rotation** around the head.

- hence the name “**cone beam**”

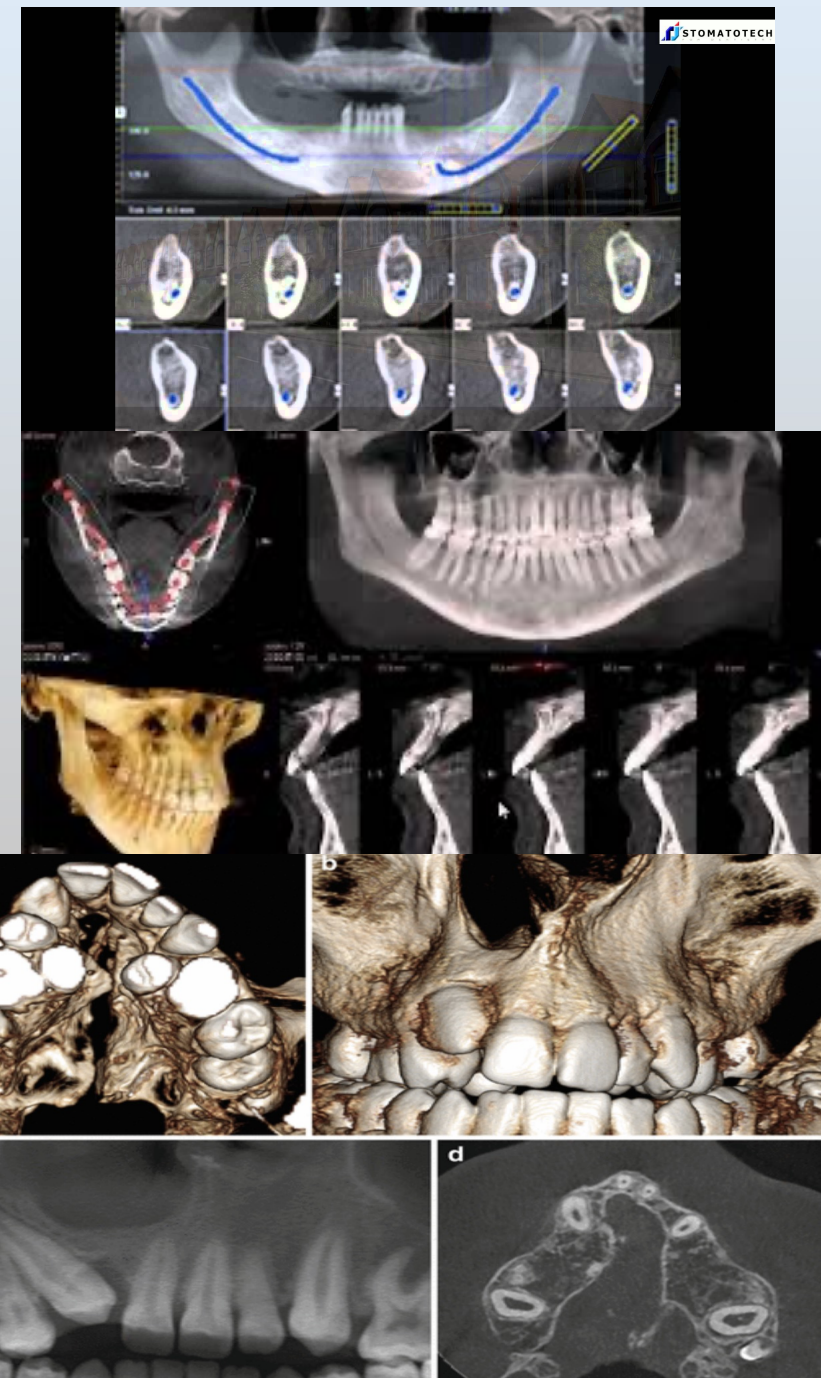
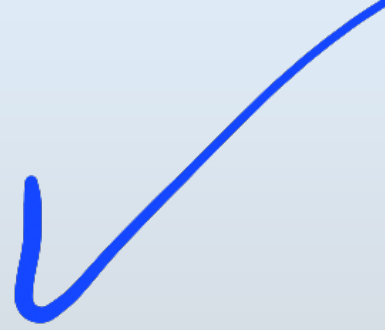
[CT uses a **linear fan beam**]



INDICATIONS:

In the assessment of :

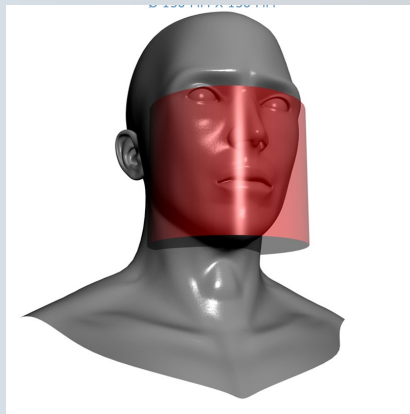
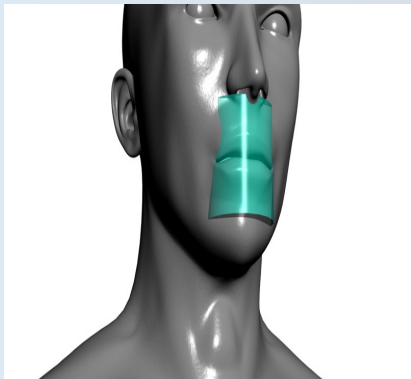
- Bony and dental pathologic conditions,
- Maxillofacial deformity ,Trauma
- Preoperative assessment of **impacted teeth**
- TMJ imaging
- In the analysis of available bone for **implant placement**.



. ADVANTAGES

Rapid scan time:. Single rotation captures all images.

- Beam limitation: Adjustable FOV(field of view) for specific needs.
- Low radiation dose: upto 40% less than CT
- Almost all the advantages of CT



• DISADVANTAGES

- Limited Soft Tissue Detail
- Radiation Exposure: lower than traditional CT scans, more than regular dental X-rays.
- Not Ideal for All Cases: -detailed soft tissue analysis (like tumor evaluation)

ULTRASONOGRAPHY

- SOUNDS are produced in the form of waves - sound waves.
- The human ear can hear frequencies between about 1,500 and 20,000 Hz.
- Diagnostic ultrasound uses vibrations in the range of 1 to 20 MHz.

ULTRASONOGRAPHY, is a diagnostic technique that uses high-frequency sound waves to produce images of structures within the body.



- The sound waves are sent into the body through a device called a **transducer**,
which also captures the echoes
that bounces back from tissues and organs

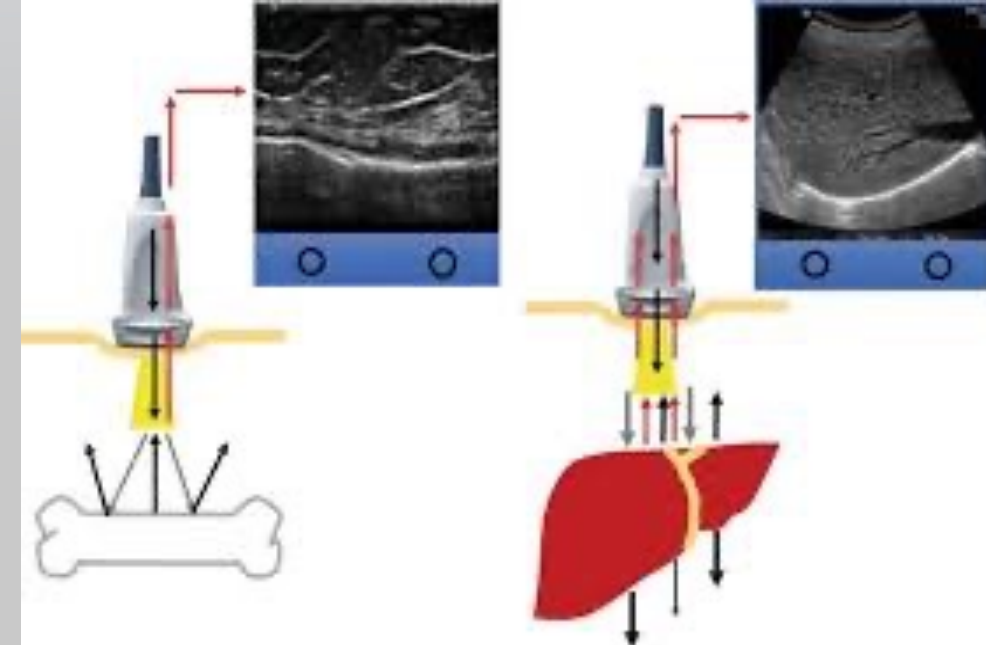
These echoes are then converted into real time images on a monitor.



TERMINOLOGIES FOR INTERPRETATION:

(based on how sound waves bounce)

- **Anechoic:** No echoes (appears dark).
- **Hyperechoic:** Strong echoes (appears bright).
- **Hypoechoic:** Weaker echoes (appears darker).



Indications In Dentistry

- Lymph node pathologies
- **Salivary gland evaluation:** detects stones, cysts, and infections in the salivary glands.
- **Muscle and joint assessment:** TMJ and facial muscles.
- **Abscess and cyst detection:** locate soft-tissue infections, abscesses, and cysts around the teeth and jaw.
- **Evaluation of oral lesions:** detects soft tissue masses and lesions, aiding early diagnosis of conditions like oral cancer.



- Advantages



- NOT ionizing radiation
- NON INVASIVE and painless imaging technique .
- Images show good differentiation between different soft tissues.
- widely available .

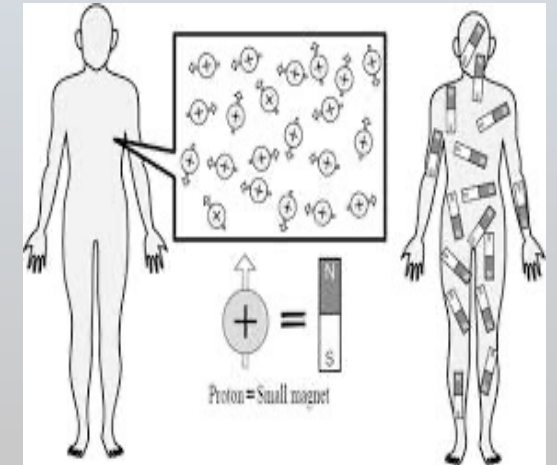
• Disadvantages



- Limited bone imaging
- Poor image resolution
- Less effective for deep lesions

MAGNETIC RESONANCE IMAGING(MRI)

- Magnetic Resonance Imaging (MRI) is a non-invasive imaging technique that uses strong magnetic fields, radio waves.
- MRI works using magnetism and hydrogen atoms (loosely bound) in the body.
- Since our body is mostly water, which contains hydrogen, each hydrogen atom has a proton that acts like a small magnet.
- When inside the MRI machine, the powerful magnet makes all the protons align with the magnetic field.

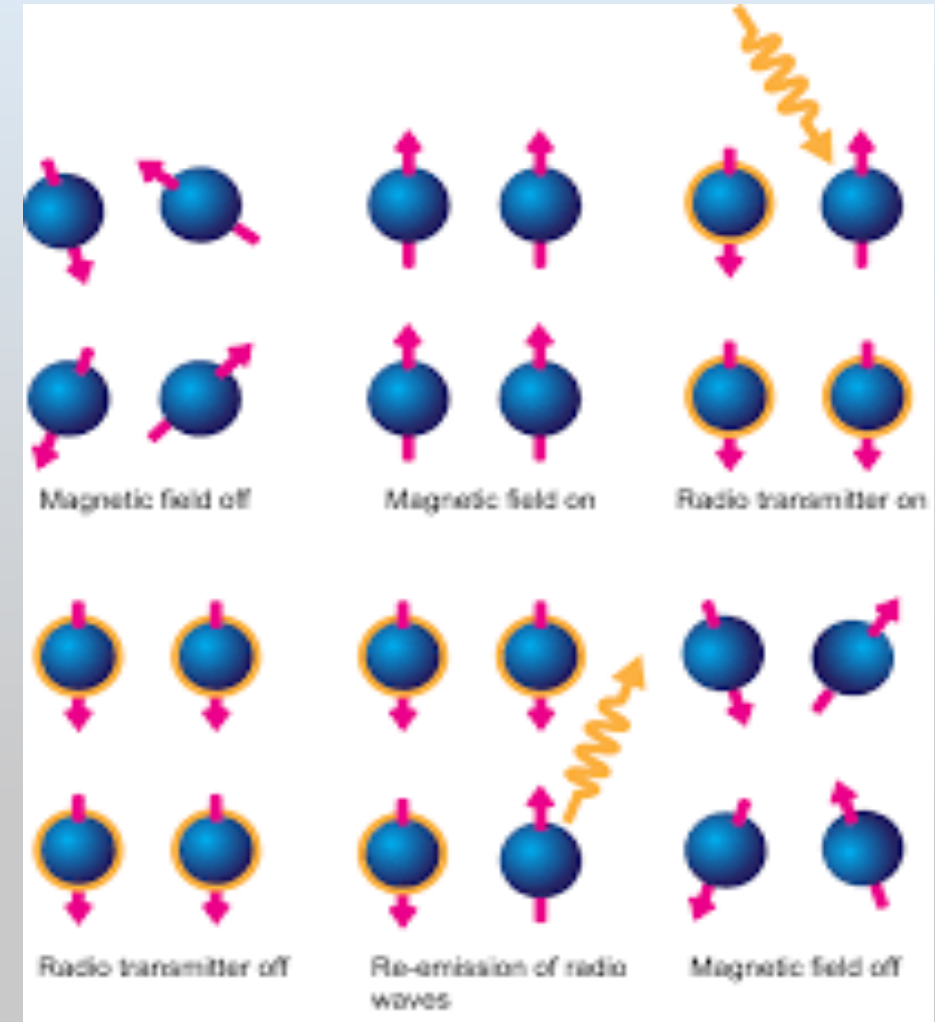


The MRI sends **radio waves** to temporarily move the protons out of alignment.

When the **radio waves** are turned off, the protons return to their original position and release energy in the form of signals.

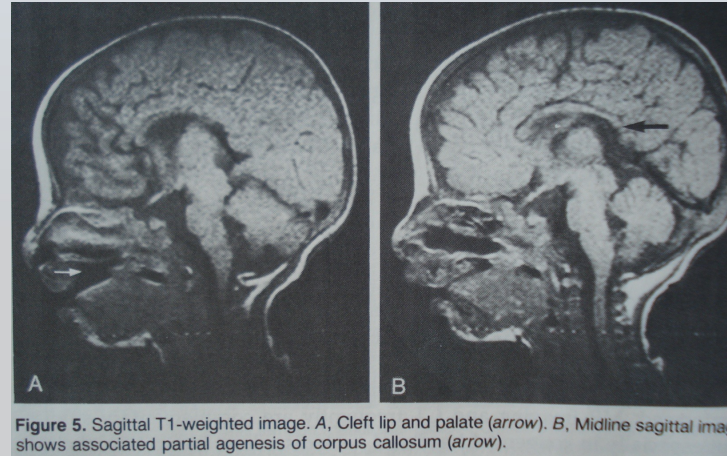
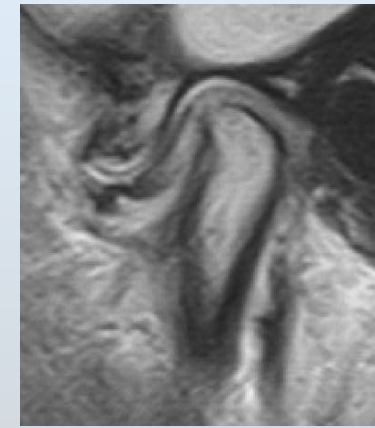
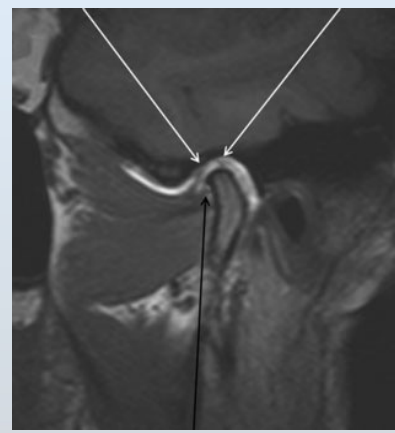
The MRI computer collects these signals and uses them to create detailed images of the body's tissues.

Appearance based on **PROTON density**.
Radiofrequency pulse sequences are used
T1 weighted image: normal anatomy
T2 weighted image to evaluate pathology



INDICATIONS

- Any Soft Tissue Pathology Of Head And Neck-cysts and tumors.
- TMJ Disc Disorders(joint space)
- Salivary Gland Disorders
- Sinusitis
- Lymph Node Pathologies



ADVANTAGES

- NON IONIZING - No radiation exposure - No biological effects
- Higher soft tissue contrast
- Blood vessels clearly visible
- Multi-planar imaging possible

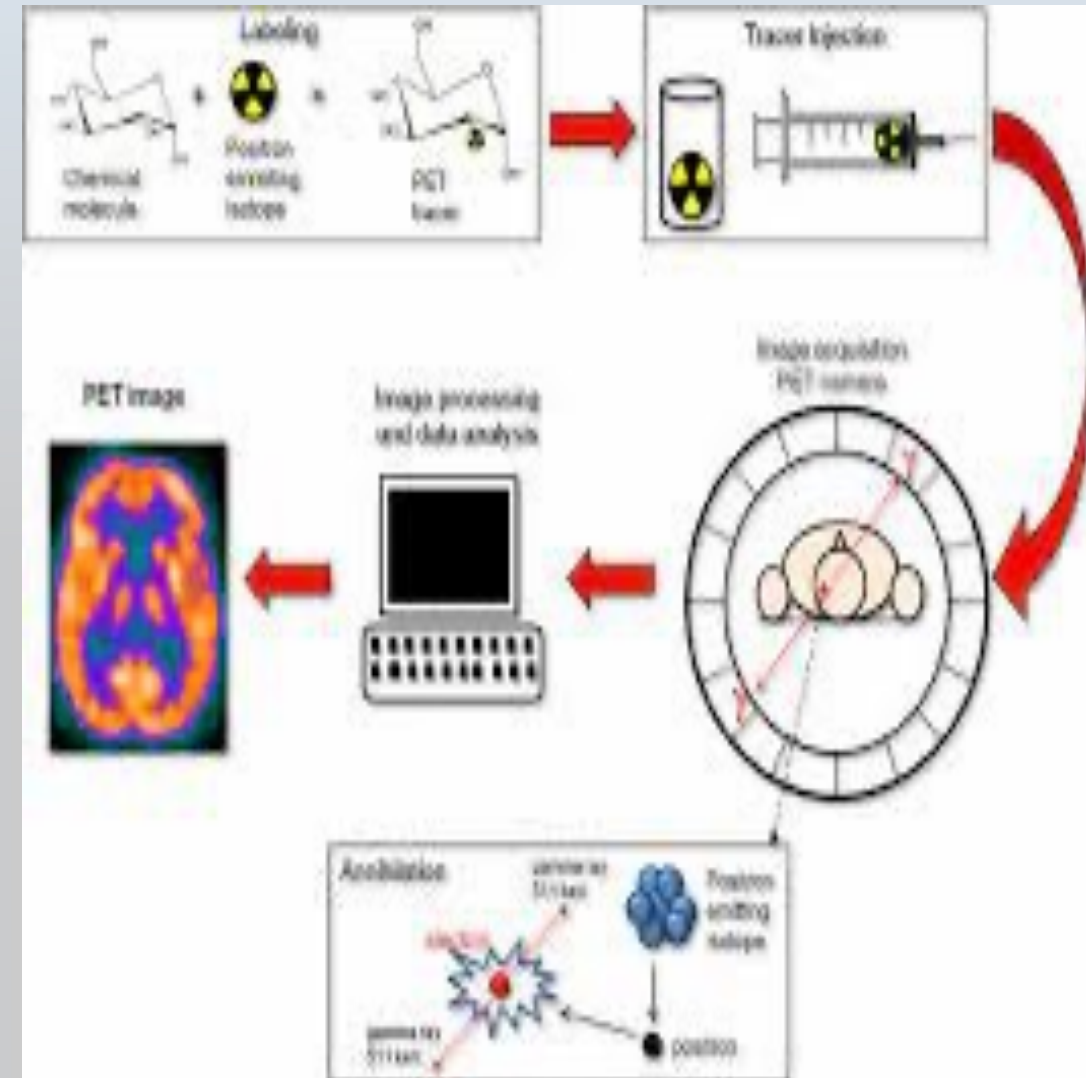
DISADVANTAGES

- Intra-bony lesions are poorly visualized
- Time consuming and inconvenience to pt.
- Claustrophobia –fear of enclosed spaces.
- Noisy
- Patients with cardiac pacemakers, and ferromagnetic implants cannot be investigated by MRI

Positron emission tomographic (PET) imaging

- PET is a nuclear imaging technique that uses a radiotracer to emit positrons.

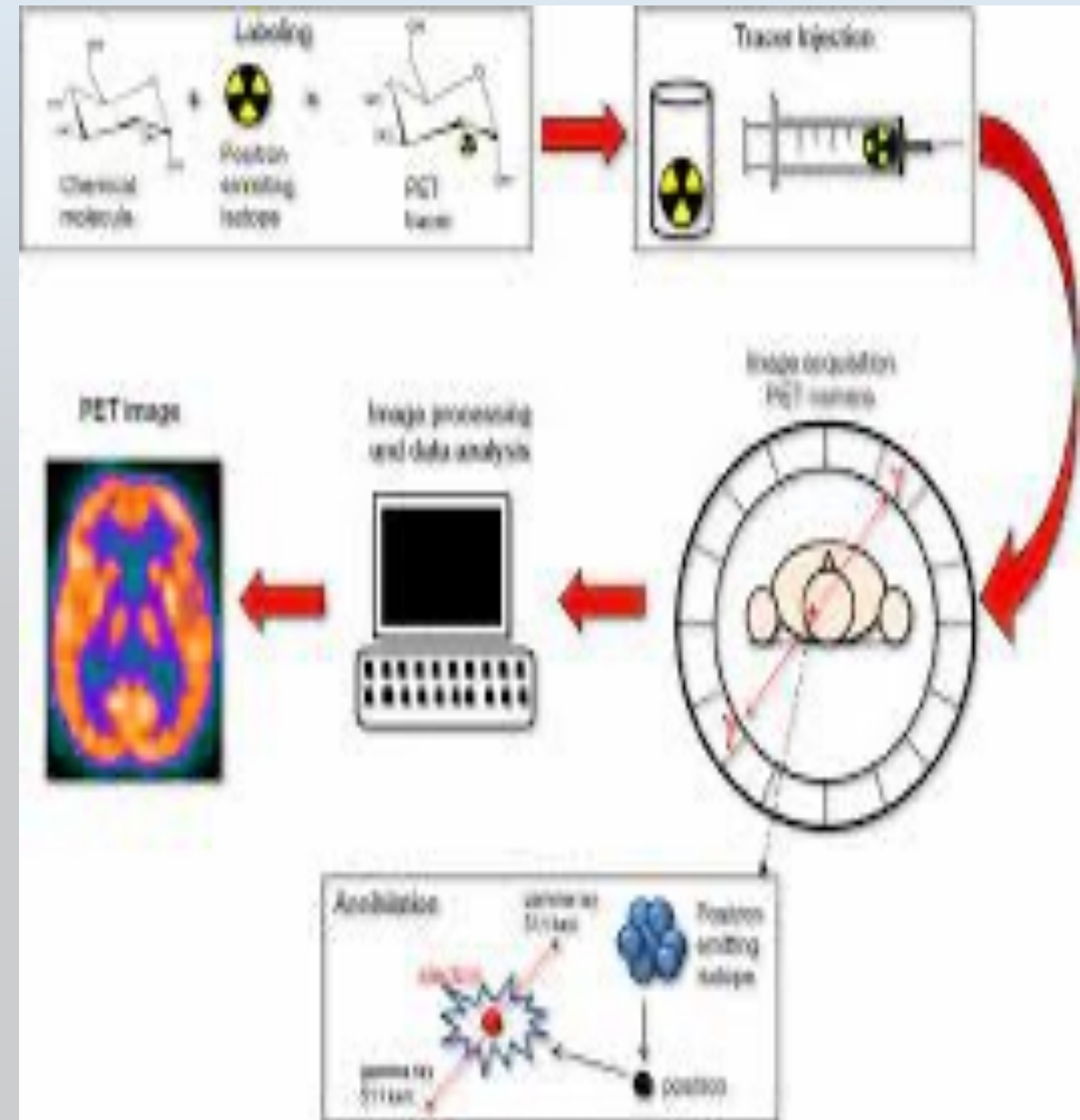
- The basic principle of a PET scan involves using a special **radiotracer (radioactive material)**. The most common tracer used is Fluorodeoxyglucose (FDG).
- When injected into the body, the tracer is absorbed by tissues, particularly those that are more active, **like cancer cells or areas of inflammation**.



The radiotracer emits **positrons**, which collide with **electrons** in the body, creating **gamma rays**.

A PET scanner(gamma camera) detects **these gamma rays** and uses them to create detailed, 3D images of how tissues and organs are **functioning**, rather than just their structure.

Positron-is a positively charged particle inside an atom ,having the same mass and magnitude of charge as the electron.



Advantages of PET:

- Provides high-resolution,
- - 3D images of metabolic and functional processes.
- Helps with early diagnosis and monitoring treatment progress.

Indications:

- Detecting and staging **cancer**, - treatment response, and monitoring for recurrence.
- Diagnosing and monitoring **neurological diseases**
- Assessing heart muscle health and blood flow.

<u>Imaging Technique</u>	<u>Typical Effective Dose (μSv)</u>
Intraoral Periapical (Single Film)	5 – 8 μSv
Full Mouth Intraoral Survey (18 Films)	90 – 150 μSv
Panoramic Radiograph (OPG)	10 – 30 μSv
Cone Beam CT (CBCT) – Small FOV	19 – 200 μSv
Cone Beam CT (CBCT) – Large FOV	60 – 600 μSv
Medical CT – Maxillofacial Region	600 – 2000 μSv
PET Scan (Whole Body)	4000 – 7000 μSv

References

- White and Pharoah, Oral Radiology- Principles and Interpretation; “**CBCT**”: **214-228 ,Edit-8**
- White and Pharoah, Oral Radiology- Principles and Interpretation; “**Other imaging modalities**” **229-249, Edit-7.**

Thank You...